

Joint continuity of the amplitude coefficients and the measurable chart integral

Claude, with Daniel Murfet

2026-09-07

Abstract

Infrastructure for the probabilistic normalised-remainder theorem (Astra #11 rank 4, second half). The amplitude coefficients $c_k(s) = \text{ampCoeff } \beta x y k s$ are finite algebraic expressions in finitely many entries of (x, y) and in s , hence jointly continuous on $\text{CoeffPair} \times \mathbb{R}$ ($\text{ampCoeff_pair_continuous}$). For a measurable random Taylor datum $X : \Omega \rightarrow \text{CoeffPair}$ the chart integral $Z_N(X(\omega))$ is therefore a measurable function of ω ($\text{measurable_chartZ_comp}$), by the parameter-measurability theorem of unit 138. The chart integral is packaged as chartZ on CoeffPair , and the uniform Taylor tree of unit 135 is restated on norm balls with the coefficient maps coeffA , coeffB ($\text{chartZ_taylor_tree_ball}$). Zero sorry/axiom.

1 The things formalised

```
theorem continuous_eval_coeffPair (i : PairIdx) : Continuous fun a : CoeffPair => a i
theorem continuous_toX / continuous_toY ( ) (k) :
  Continuous fun a : CoeffPair => toX a k
theorem convPow_continuous_param (f : X → × → )
  (hf : k, Continuous fun p => f p k) (n) :
  k, Continuous fun p => convPow (f p) n k
theorem dropConst_continuous_param / expCoeff_continuous_param
theorem ampCoeff_continuous_param ( ) (x y : X → × → ) (hx hy) (s : X → ) (hs) (k) :
  Continuous fun p => ampCoeff (x p) (y p) k (s p)
theorem ampCoeff_pair_continuous ( ) (k) :
  Continuous fun q : CoeffPair × => ampCoeff (toX q.1) (toY q.1) k q.2
theorem ampCoeff_pair_measurable (X : Ω → CoeffPair) (hX : Measurable X) (k) :
  Measurable fun p : Ω × => ampCoeff (toX (X p.1)) (toY (X p.1)) k p.2
def chartZ ( b ) (h h k k) (N) (a : CoeffPair) : :=
  twoDamp b N h h k k (anaAmp (ampCoeff (toX a) (toY a)) b)
theorem measurable_chartZ_comp ... (X) (hX : Measurable X) :
  Measurable fun => chartZ ... N (X )
theorem measurable_chartZ ... : Measurable (chartZ b h h k k N)
theorem chartZ_taylor_tree_ball ... (M) (hM : 0 M) (a) (ha : ||a|| M) (T N)
  (hN : 1 N) :
  |chartZ ... N a - polesBelowGen h h k k p p T,
  N^(-) * (coeffA ... a * log N + coeffB b ... a)|
  uniformTreeConst b M (2 * M) p p T h h k k 0 * (N^(-(2 * T)) * (1 + log N))
```

2 Mathematics

Each $c_k(s)$ is $e^{\beta s x_{00}} \sum_{a \leq k} y_a E_{k-a}(\beta s)$ with $E_m(t) = \sum_{n \leq |m|} \frac{t^n}{n!} (\bar{x}^{*n})_m$, and $(\bar{x}^{*n})_m$ is a finite sum of products of entries of $\bar{x}$; so c_k is a polynomial in finitely many entries of (x, y) and in $e^{\beta s x_{00}}$, s . Point evaluation is 1-Lipschitz on ℓ^1 , so all of this is jointly continuous on $\mathbf{CoeffPair} \times \mathbb{R}$, hence Borel measurable (the product of a metric space with the second-countable $\mathbb{R}$ has Borel σ -algebra equal to the product σ -algebra). Composing with a measurable X gives the joint measurability hypothesis of unit 138's parameter theorem, whose conclusion is the measurability of $\omega \mapsto Z_N(X(\omega))$. On the ball $\|a\| \leq M$ the amplitude has the common envelope $M e^{2\beta M s}$ (unit 145), so the uniform Taylor tree applies with $C_0 = M$, $L = 2M$, $D = 0$.

3 Paper-facing meaning

Two of the three ingredients of the probabilistic normalised-remainder statement are now in place: the deterministic bounds (unit 148, applied with $K = \mathbf{uniformTreeConst}$ from `chartZ_taylor_tree_ball`) and the measurability of the random chart integral. The next unit assembles them with Slutsky's theorem: for random Taylor data bounded by M and converging in distribution, $(Z_n - L_{<\alpha}) / (N_n^{-\alpha} \log N_n) \Rightarrow A_\alpha(Z)$ and the B-slot analogue.

4 Lean notes

- Continuity by induction on the convolution power: the `zero` case is `change Continuous fun _ => delta k` (definitional unfolding of `convPow a 0`), the `succ` case unfolds `conv` to a `continuous_finsetSum`.
- Binders over a product must be type-annotated (`fun q : CoeffPair × => ...`); otherwise projections `q.1` fail to elaborate (“Invalid projection”).
- `lp.lipschitzWith_one_eval 1 i` gives continuity of point evaluation on ℓ^1 ; `Continuous.measurable` on `CoeffPair ×` finds the product `OpensMeasurableSpace` instance through `SecondCountableTopologyEither` with the $\mathbb{R}$ factor.
- The file compiled after one binder-annotation fix.